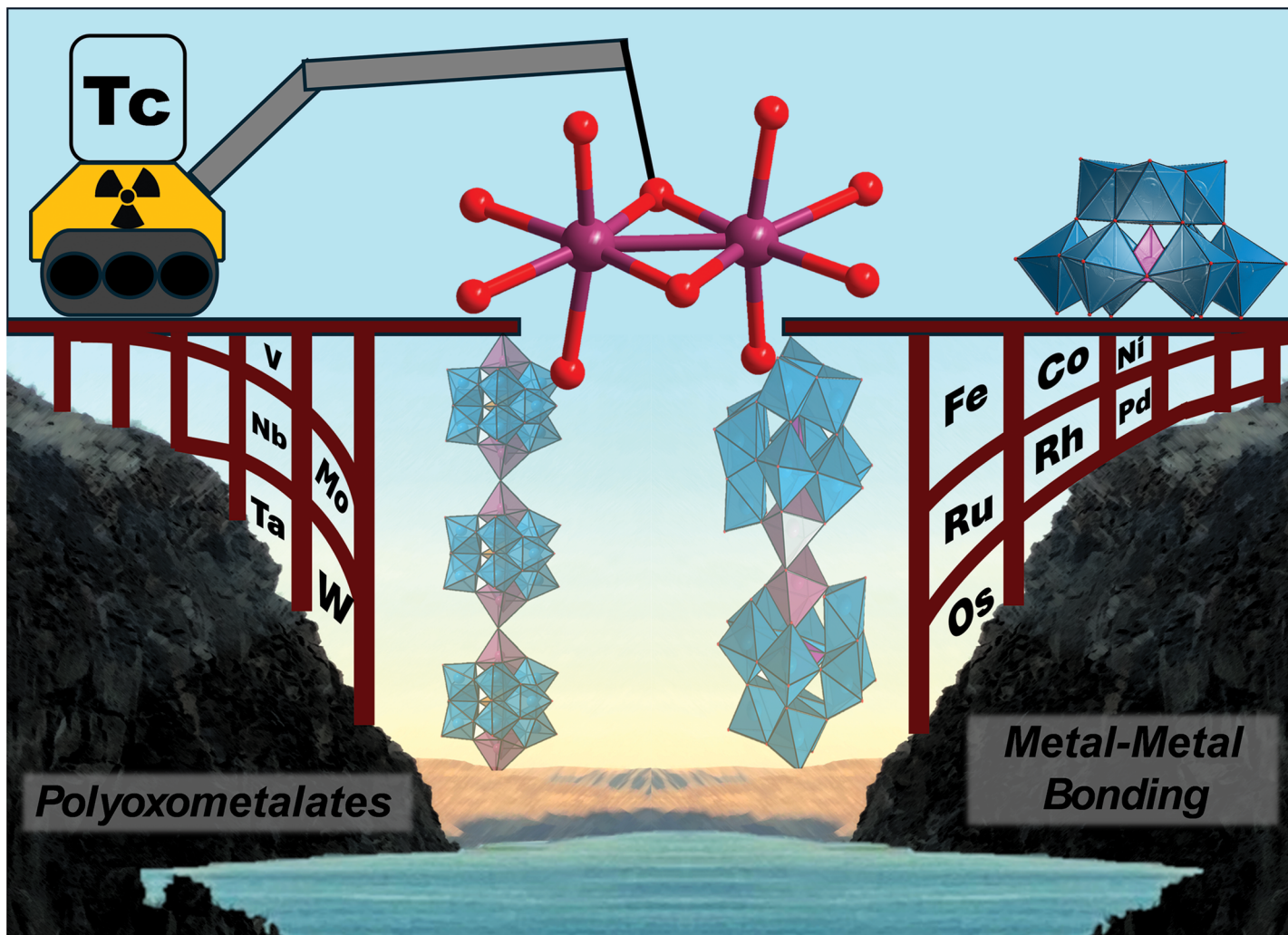




Showcasing research from Professor May Nyman's laboratory, Department of Chemistry, Oregon State University, Corvallis, USA

Building block approach to technetium-substituted polyoxotungstates

Group 7 technetium has long exhibited classic metal-metal bonded dimer/cluster chemistry, in addition to emergent polyoxometalate-like behaviour. In the first two crystal structures of Tc-substituted polyoxometalates, Tc demonstrates hybrid bonding of both metal-metal bonded complexes and polyoxometalates that bridge the middle of the periodic table.

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Building block approach to technetium-substituted polyoxotungstates

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Single-crystal X-ray structures of $\text{Na}_{14}[\text{Na}_2(\text{H}_2\text{O})_4\text{Tc}_2\text{O}_2(\alpha\text{-SiW}_{10}\text{O}_{37})_2]\cdot 35\text{H}_2\text{O}$ (Tc-POM-dimer) and $\text{Cs}_7(\text{Tc}^{\text{VI}}\text{O}_4)\text{trans-}[\text{Tc}_2^{\text{IV}}\text{PW}_{10}\text{O}_{39}]\cdot 9\text{H}_2\text{O}$ (Tc-POM-chain) are the first X-ray structures for Tc-substituted polyoxometalates. Lacunary polyoxometalates accommodating Tc-dimers, impose asymmetry in crystal lattices without Tc/W disorder. Computation describes putative $\text{Tc}^{\text{VI}}\text{O}_4^{2-}$ in Tc-POM-chain and explains Tc-connectivity in Tc-POM-dimer, which has characteristics of both POMs and metal-bonded dimers.

Polyoxometalates (POMs) are traditionally defined as d^0 group 5 and group 6 polynuclear, anionic oxoclusters. POMs are distinguished from other oxoclusters by the multiply-bonded η -oxos ligands that cap and stabilize them. The organic ligands that stabilize other oxoclusters exclude their description in calculated Pourbaix and speciation diagrams. The η -oxos enable (i) precisely defined nuclearity and structure, (ii) reactivity unencumbered by ligands, and (iii) high solubility in both organic solvents and water, controlled by counter cations and anionic charge.¹ Chemistry and function of POMs are further diversified with formation of lacunary derivatives (missing polyhedra) and substitution of open-shell transition metals, main group metals, and f-elements into the *lacunas*. One relevant POM application is radioisotope isolation and separation that has been demonstrated extensively with actinides.² In this function, the purported radiation stability of the all-inorganic ligands and the multi-denticity of lacunary POMs are desirable characteristics. Innovations leveraging POM-radionuclide complexation has inspired new ideas for nuclear fuel reprocessing, nuclear waste disposal, and metal recovery.² Equally important, fundamental understanding of bonding metrics and trends,³ electronic configurations,⁴ and f-element

absorption/emission⁵ at the frontier of the periodic table is enabled by POM-transuranic complexation and crystallization. Complementarily, these studies provide new insight into POM behaviour including isomerization in solution,⁶ elucidated by complexation of actinides.

Comparably informative as POM-actinide complexation but far less-studied is POM-technetium complexation. Technetium (^{99}Tc) is a high yield decay product of uranium/plutonium nuclear materials. Due to facile oxidation state change of technetium that drives coordination, precipitation and dissolution; ^{99}Tc is problematic in nuclear waste handling and nuclear fuel reprocessing.⁷ In addition, $^{99\text{m}}\text{Tc}$ (m = metastable) underpins a multi-billion dollar radiopharmaceutical industry for both imaging and therapy. Its effective production and separation from the $^{99}\text{Mo}_7\text{O}_{24}^{6-}$ -POM also requires understanding and efficient control over oxidation state and speciation.⁸ In both these technologically relevant settings, the high radioactivity enhances Tc redox lability which can range from Tc^{III} to Tc^{VII} in aqueous conditions, challenging predictable behaviour.

With a periodic table position just to the right of POM-forming metals, Tc (and heavier congener Re) exhibits POM-like character, including stabilization by the η -oxo in high oxidation states ($\text{Tc}^{\text{V/VI/VII}}$). Recently, the first Tc-POM⁹ and mixed Tc/Re-POM¹⁰ were reported, suggesting POM chemistry could be expanded to group 7. Despite these similarities, even studies of Tc/Re-substituted group 6 (tungstate) POM complexes have been relatively few; and Tc/Re-substitution into Mo, V, Nb and Ta POMs is non-existent. Challenges to expanding Tc/Re-substituted POMs includes control over Tc/Re oxidation state and crystallography. Progress to date is reviewed below.

Pope and Francesconi respectively provided the first studies of $\text{Re}^{\text{IV/V/VI}}$ and $(\text{Re/Tc})^{\text{IV/V/VI}}$ substituted W-POMs. Pope described cyclic voltammetry (CV) and electron spin resonance of Re-substituted lacunary Keggin $\text{XW}_{11}\text{O}_{39}$ ($\text{X} = \text{B}, \text{Si}, \text{P}$).¹¹ Pope also reported the single crystal X-ray structure (SCXRD) of $[\text{Re}^{\text{V}}\text{W}_9\text{O}_{32}]^{5-}$ with Re^{V} disordered over eight sites.¹² Francesconi performed Tc-substitution into the monolacunary Wells Dawson ($\text{P}_2\text{W}_{17}\text{O}_{61}^{10-}$) POM¹³ and the monolacunary Keggin POM.¹⁴

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Multinuclear NMR, Tc K-edge X-ray absorption spectroscopy (XAS), and CV concluded mono-substitution of $\text{Tc}^{\text{V}}=\text{O}$ into the POMs. In additional studies, Francesconi proposed multiple possible structural models of Tc-POM complexes from solution data¹⁵ including $\text{Tc}^{\text{V}}=\text{O}$ substitution within the lacunary POMs, $\text{Tc}_2^{\text{IV}}\text{O}_2$ coordinated on the surface of plenary POMs, and polymeric $\text{Tc}^{\text{IV}}\text{O}_2 \cdot 2\text{H}_2\text{O}$ stabilized by non-covalent association with the POMs.¹⁶ No SCXRD data were obtained to confirm or rule out models.

In fact, no Tc-substituted POMs have been structurally characterized and only three crystal structures of Re-substituted POMs without disorder have been reported. Kato¹⁷ isolated $[\text{O}(\text{Re}^{\text{V}}(\text{OH})[\text{P}_2\text{W}_{17}\text{O}_{61}])_2]^{14-}$ from the reaction of $[\text{P}_2\text{W}_{17}\text{O}_{61}]^{10-}$ and $[\text{Re}^{\text{IV}}\text{Cl}_6]^{2-}$, where a Re_2O dimer sits outside of the lacunary holes of the two flanking POMs. In a second example, Neumann crystallized $[\text{Re}^{\text{VI}}_3\text{P}_2\text{W}_{15}\text{O}_{62}]^{5-}$ from $[\text{P}_2\text{W}_{15}\text{O}_{56}]^{12-}$ and $[\text{Re}^{\text{VO}}\text{Cl}_3(\text{PPh}_3)_2]^{18}$. Third, Sokolov¹⁹ crystallized $[\text{Re}_2(\text{PW}_{11}\text{O}_{39})_2]^{8-}$, with a $\text{Re}^{\text{III}}-\text{Re}^{\text{III}}$ quadruply-bonded dimer bridging the two POMs, from $(\text{PW}_{11}\text{O}_{39})^{7-}$ and $[\text{Re}^{\text{IV}}_2\text{Cl}_8]^{2-}$. These three studies suggest that while the lacunary POM does not change in the reaction, neither the nuclearity nor the oxidation state of the group 7 starting material necessarily dictates its speciation within the POM. These studies also imply that unique coordination of the substituted group 7 metal *via* dimer or trimer formation enables structural differentiation from tungsten. Otherwise, the nearly identical number of electrons for W and Re makes then indistinguishable by SCXRD. Despite the greater difference between W and Tc electron count, SCXRD data is completely lacking for Tc-substituted POMs, and molecular level technetium oxide bonding information is also sparse.

SCXRD data for Tc-O molecules including Tc-substituted POMs would enable (1) Reliable bond valence sum (BVS) parameters for Tc-O in different oxidation states, (2) Accurate models for computation, (3) Informed interpretation of Tc-O solutions, and (4) Understanding technetium in technologically-relevant matrices including those for $^{99\text{m}}\text{Tc}$ production and ^{99}Tc in nuclear wastes. Here we exploit labile tri-vacant lacunary Keggin ions that can accommodate polynuclear Tc-species, rather than *vice versa* with more rigid POMs. We crystallized the Tc starting material plus two Tc-POM complexes, demonstrating preservation of the oxidation state (Tc^{IV}) and the nuclearity (dimer) of the Tc starting material, while the W-POM undergoes rearrangement. Despite non-ideal diffraction data with challenges including twinning, weak diffraction at high angles caused by lattice disorder, and absorption challenges of W; the presented X-ray structures accurately depict the Tc-substituted POM arrangements. Additional details are provided in the supporting information plus Fig. S1–S3. Both $\text{Na}_{14}[\text{Na}_2(\text{H}_2\text{O})_4\text{Tc}_2\text{O}_2(\alpha\text{-SiW}_{10}\text{O}_{37})_2] \cdot 35\text{H}_2\text{O}$ (**Tc-POM-dimer**) and $\text{Cs}_7(\text{Tc}^{\text{IV}}\text{O}_4) \text{ trans-}[\text{Tc}^{\text{IV}}\text{PW}_{10}\text{O}_{39}] \cdot 9\text{H}_2\text{O}$ (**Tc-POM-chain**) feature a Tc_2^{IV} oxo-bridged dimer, similar to the starting material, $[\text{Tc}_2\text{OCl}_{10}]^{4-}$ (Fig. 1).

Dissolution of $\text{H}_4\text{Tc}_{20}\text{O}_{68}$ ($\text{Tc}_{20}\text{-POM}^9$) in dilute HCl produces a green mixed $[\text{Tc}^{\text{IV}}_2\text{OCl}_{10}]^{4-}/[\text{Tc}^{\text{VII}}\text{O}_4]^-$ solution (Fig. S4). UV-vis absorption peaks (Fig. S4) at 244 nm and 287 nm are characteristic of TcO_4^- , and the peaks at 358 nm, and 607 nm are attributed to $[\text{Tc}_2\text{OCl}_{10}]^{4-}$. The $[\text{Tc}_2\text{OCl}_{10}]^{4-}$ anion

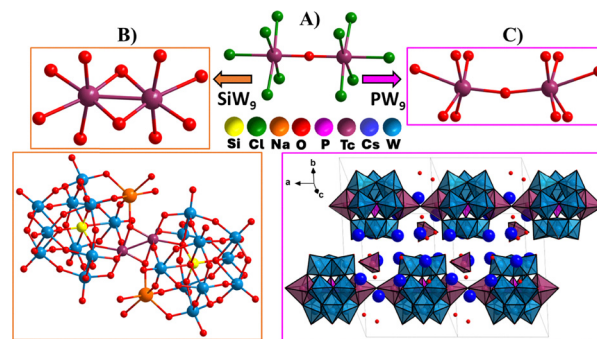


Fig. 1 Overview of synthesis of **Tc-POM-dimer** (left) and **Tc-POM-chain** (right) (A) $[\text{Tc}_2\text{OCl}_{10}]^{4-}$, (B) Tc_2O_2 core of **Tc-POM-dimer** (top) and **Tc-POM-dimer** (bottom) (C) Tc_2O bridge of **Tc-POM-chain** (top) and **Tc-POM-chain** lattice view (bottom).

was proposed as a hydrolysis and photodegradation product of $[\text{TcCl}_6]^{2-}$ 20 yrs prior,²⁰ but never isolated. We isolated $[\text{Tc}_2\text{OCl}_{10}]^{4-}$ as $\text{K}_4\text{Tc}_2\text{OCl}_{10}$ by addition of KCl. $\text{K}_4\text{Tc}_2\text{OCl}_{10}$ crystallizes in the tetragonal space group $I4/mmm$ (Table S1). The structure consists of a $\text{Tc}_2^{\text{IV}}\text{O}_2$ core with a Tc–O–Tc bond angle of 180° , Tc–O bond distance of $1.850(4)$ Å, and five Tc–Cl bond distances of from $2.380(4)$ Å to $2.385(6)$ Å (Fig. 1a). The four K^+ counteranions have K–Cl bond distances ranging from $3.164(4)$ – $3.516(6)$ Å (Fig. S5). Charge balance and BVS confirm Tc^{IV} oxidation state (Table S2). Isostructural Re, Ru, and W analogues have been reported.²¹ DFT calculations for $[\text{Tc}_2\text{OCl}_{10}]^{4-}$ and $\text{K}_4\text{Tc}_2\text{OCl}_{10}$ (Table S3) exhibit clearly that the open-shell singlet is more stable than triplet and septet electronic states, suggesting one unpaired electron per Tc^{IV} . The experimental Tc–O distance also matches well that of the singlet state (1.88 Å calculated; 1.85 Å experimental). **Tc-POM-dimer** was isolated by adding the $[\text{Tc}_2\text{OCl}_{10}]^{4-}$ solution to buffered $\text{Na}_{10}\text{SiW}_9\text{O}_{34}$ (SiW_9 , details in SI). After 5 days, pink rhombohedral crystals (Fig. S6) are deposited. **Tc-POM-dimer** crystallizes in the triclinic space group $P\bar{1}$ (Table S1), formulated $\text{Na}_{14}[\text{Na}_2(\text{H}_2\text{O})_4\text{Tc}_2\text{O}_2(\alpha\text{-SiW}_{10}\text{O}_{37})_2] \cdot 35\text{H}_2\text{O}$. The polyanion consists of a $[\text{Tc}_2\text{O}_2]^{4+}$ core (Fig. 1b) bridging two *cis*-divacant, SiW_{10} Keggin units (Fig. 1c). We cannot definitively state that $[\text{Tc}_2^{\text{IV}}\text{O}_2]^{4+}$ originates from the $[\text{Tc}_2\text{OCl}_{10}]^{4-}$ and not $[\text{TcO}_4]^-$, but seems the likely reaction path, given the identical oxidation state and nuclearity. Each Tc atom is bound to five O^{2-} -ligands of SiW_{10} , one bridging to the central SiO_4 , three bridging to W^{VI} , and one bridging to Na^+ , with $\text{Tc-O}_{\text{Keggin}} = 1.949$ – 2.181 Å (Table S4). The 6-coordinate Na^+ cation bridged to Tc resides in the second void of the divacant Keggin ion ($\text{Na-O} = 2.26$ – 3.10 Å, Table S4). Tc is 7-coordinate, with a Tc–Tc distance of $2.398(4)$ Å, and the Tc– μ -O–Tc bond angle is $76.9(7)^\circ$.

Interestingly, a prior XAS study comparing Tc^{IV} species in the presence of $\text{PW}_{12}\text{O}_{40}^{3-}$ to $\text{SiW}_{12}\text{O}_{40}^{4-}$ ¹⁵ suggested $\text{Tc}_2^{\text{IV}}\text{O}_2$ is stabilized by coordinating to the outside of plenary $\text{SiW}_{12}\text{O}_{40}^{4-}$, whereas $\text{PW}_{12}\text{O}_{40}^{3-}$ is a weaker ligand, and $\text{Tc}^{\text{IV}}\text{O}_2$ forms instead. This prior study provided insight into Tc-POM complexes, but it did not have the benefit of a crystal structure and also did not consider higher lability of silicotungstate and more ready formation of lacunary derivatives.

Tc-POM-dimer is charge-balanced by 14 Na⁺ that all are located on fully-occupied sites within the crystal lattice. Most notable is Na1 that links the POM-dimer into infinite chains along the *c*-direction (Fig. S7). Na1 is anti-prismatic, with four Na–O bonds (~2.6 – 2.7 Å) to each neighbouring POM. The chains stack along the *a*-direction (Fig. S8) and are further connected by sodium cations within the *ab*-plane. The [Tc₂O₂]⁴⁺ unit has been previously crystallized using multidentate organic ligands (Table S5) with Tc–Tc bond distances ranging from 2.33 to 2.40 Å (Table S5), compared to 2.398 Å for **Tc-POM-dimer**. We used DFT methods to describe the molecular orbitals (Fig. 2a) responsible for both Tc–O_{POM} and Tc–Tc bonds, as well as the bond order. Instead of the anticipated $\sigma^2\pi^4\delta^0$ electron configuration of a classic metal–metal bond, theory predicts a $\sigma^2\pi^2\delta^2$ electron configuration. The Tc–Tc δ -bond (HOMO), exhibits barely overlapping atomic orbitals, explaining the long bond. Moreover, the empty Tc-*d_{xy}* orbitals participate in the Tc₂O₂ central scaffold of the molecule, accounting for this diversion from the conventional molecular orbital theory developed decades prior.²² The three Tc–Tc bonds include: (i) σ -bond from the Tc *d_{x²-y²}*-like orbitals (HOMO–2). For classic M₂-dimer structures, the *d_{z²}* orbitals usually provide the M–M σ -bond. (ii) π -bond from the Tc *d_{yz}* orbitals (HOMO–1), and (iii) δ -bond from the Tc *d_{xz}* orbitals (HOMO). In classic M₂-dimer structures, the *d_{xz}* orbitals usually form a π -bond. In the prior-reported Tc₂^{IV}O₂ ligated with EDTA,²³ the authors propose a $\sigma^2\pi^2\delta^2$ electron configuration based on calculations. These differences suggest bonding to the POM influences the bonding within the Tc₂O₂ unit. Computation predicts nearly identical energy for singlet and triplet states (Table S6), but the bond length of the singlet state agrees with the experimentally observed bond length, suggesting antiferromagnetic coupling of Tc's valence electrons across the di-oxo bridge.

Addition of a [Tc₂OCl₁₀]^{4–} solution to SiW₉ results in a colour change from orange to purple, tracked *via* UV-vis spectrophotometry (Fig. 2b). Absorption at 540 nm grows and red-shifts and stabilizes at 562 nm after one hour, accounting for the colour change from orange to purple. This change is interpreted as the transition from a Tc–O–Tc species to a Tc– μ O₂–Tc species.¹⁵ Additionally, upon introducing the [Tc₂OCl₁₀]^{4–} solution to the POM solution, the absorptions characteristic of [Tc₂OCl₁₀]^{4–} (Fig. S4) disappear, indicating both hydrolysis of the oxychloride species at pH = 5, and replacement of the chloride ligands by POM ligation. The SiW₉ solution exhibits absorbance bands in the UV-region (Fig. S9), and with addition of the [Tc₂OCl₁₀]^{4–} species, the absorbance red-shifts, indicating complexation of Tc to silicotungstate.¹⁵ Additional indirect evidence of Tc-POM complexation is that TcO₂·nH₂O precipitates in the absence of the POM that replaces labile chloride ligands.

Small-angle X-ray scattering (SAXS) of the **Tc-POM-dimer** reaction solution is compared to SiW₉ without added Tc, plus simulated scattering curves for SiW₉, SiW₁₁ and **Tc-POM-dimer** (Fig. 2c). The SiW₉ experimental scattering alone without added Tc (green curve) is similar to the simulation for either SiW₉ or SiW₁₁. The **Tc-POM-dimer** reaction solution (red curve) is shifted to lower *q*-range in the Guinier region (ranges from 0.1 to 0.7 Å^{–1}) compared to simulated SiW₉ and SiW₁₁ and closely resembles the simulated **Tc-POM-dimer** scattering. The existence of dimers is also evidenced by the pair distance distribution function analysis (PDDF, a histogram of probability vs scattering length through the clusters). The PDDF (Fig. S10) features two Gaussian peaks, with the second being less intense than the first, characteristic of a dimer. The radius of gyration (*R_g*) obtained from PDDF modeling is 5.40 Å (Fig. S10), smaller than *R_g* of simulated **Tc-POM-dimer** SAXS (5.87 Å), indicating the presence of some monomers, *i.e.* [SiW₁₁TcO₄₀]. This latter complex was tentatively identified within other crystals, but the data quality was too poor to report, perhaps supporting the inability to access reliable SCXRD structures in prior studies by Pope and Francesconi. Obtaining mixed Tc-substituted POMs is not surprising, given the lability of the silicotungstate POMs.

Tc-POM-chain from [PW₉O₃₄]^{9–} and [Tc₂OCl₁₀]^{4–} crystallizes in the monoclinic space group *P*₂₁/*m*, fully formulated Cs₇[Tc₂^{IV}O(PW₁₀O₃₈)(Tc^{VI}O₄)]·9H₂O (Tables S1 and S7). In this structure, there are two different Tc species; Tc^{IV}₂O likely originates from the [Tc₂OCl₁₀]^{4–} moiety, while putative [Tc^{VI}O₄]^{2–} is most likely reduced from [TcO₄][–] in the reaction solution. Electrons for [TcO₄][–] reduction could originate from Cl[–] or Tc^{IV}. The linear Tc₂O unit connects *trans*-divacant PW₁₀ Keggin units into chains along the *a*-axis (Fig. 1c). The Tc₂O unit, isostructural with [Tc^{IV}₂OCl₁₀]^{4–} starting material, exhibits a Tc–Tc distance of 4.200(5) Å and Tc–O bond distances of 1.995(7) Å and 2.256(7) Å. The five Tc–O_{POM} distances range from ~1.980–2.460 Å. The Tc– μ –O–Tc bond is bent at 162.2(2)°, attributed to coordination with two Cs-cations, Fig. S11a. Seven charge-balancing Cs⁺, nine water molecules and a pseudo-tetrahedral TcO₄ anion (one per Keggin ion, Fig. S11b) reside between the chains. Computational studies considering discrete models of **Tc-POM-chain** were carried out for theoretical

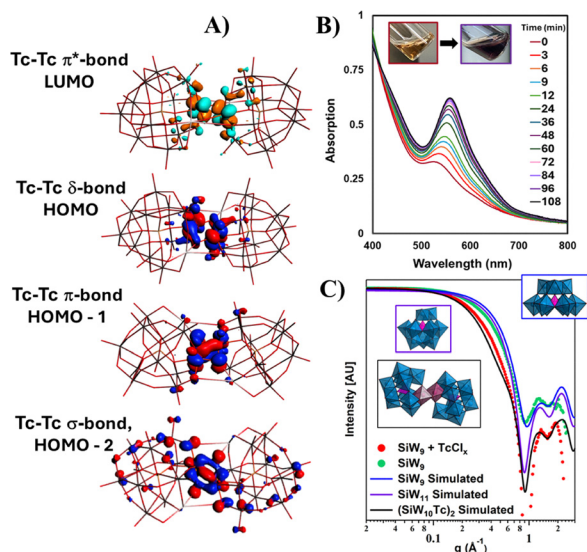


Fig. 2 (A) computed frontier orbitals for **Tc-POM-dimer**. (B) UV-vis spectra of the transition of **Tc-POM-dimer** reaction solution and (C) SAXS of SiW₉ with (red) and without (green) added Tc. Simulated SAXS includes Tc-POM-dimer (black), SiW₉ (blue) and SiW₁₁ (purple).

characterization. These models consist of two oxo-bridged [Te_2W_{10}] Keggin-type units, with the terminal oxos saturated with hydrogen atoms and fixed during geometry optimization to account for the strain imposed by the extended chain (Fig. S12 and S13). The molecular geometry of the Tc-O-Tc moiety is strongly affected by the oxidation state, the electronic state and the model itself. Our most reliable DFT calculations predict an open-shell singlet ground state, in which the unpaired $d(\text{Tc})$ electrons exhibit antiferromagnetic coupling across the Tc-O-Tc bridges (Tables S8–S10 and Fig. S14) as shown by the spin density maps. Notably, the inclusion of Cs^+ counteranions in the models was required to reproduce the geometric parameters around the $[\text{Tc}^{\text{IV}}_2\text{O}]$ motif.

Finally, we analyzed the TcO_4 -oxoanion by DFT. Based on both charge-balance requirements of the framework and the geometry of this anion, we consider the possibility of $\text{Tc}^{\text{VI}}\text{O}_4^{2-}$ instead of common $\text{Tc}^{\text{VII}}\text{O}_4^-$. The long bonds and angular distortion from ideal tetrahedral geometry were considered. The bond distances support Tc^{VI} convincingly, while the bond angles provide less certain evidence. Fig. S15 summarizes and compares the experimental and computed metrics for TcO_4 within the **Tc-POM-chain** lattice to Tc-O bond lengths and O-Tc-O bond angles taken from the approximately 30^{24} pertechnetate structures in the CCDC/ICSD databases. We note that the calculated energy penalty for distorting a $\text{Tc}^{\text{VII}}\text{O}_4^-$ anion from T_d to C_{2v} , or a $\text{Tc}^{\text{VI}}\text{O}_4^{2-}$ anion from C_{2v} to T_d is small, $<0.5 \text{ kcal mol}^{-1}$ (Tables S11 and S12). Also, DFT calculations on imposing either $\text{Tc}_{\text{POM}}^{\text{IV}}/\text{Tc}^{\text{VI}}\text{O}_4^{2-}$ or $\text{Tc}^{\text{III}}/\text{Tc}_{\text{POM}}^{\text{IV}}/\text{Tc}^{\text{VII}}\text{O}_4^-$ to match charge-balance yielded structural parameters that are ambivalent to oxidation state; both agree equally well with the structural data (Tables S8–S10). Meanwhile, BVS calculations (Table S7) of the experimental data better support the proposed $\text{Tc}_{\text{POM}}^{\text{IV}}/\text{Tc}^{\text{VI}}\text{O}_4^{2-}$ technetium oxidation states. The $[\text{M}^{\text{VI}}\text{O}_4]^{2-}$ moiety was proposed prior for Mn, Tc and Re, unfortunately without single-crystal description.²⁵

This study reports the first crystal structures of Tc-substituted POMs, and tentatively the first structural evidence for $\text{Tc}^{\text{VI}}\text{O}_4^{2-}$. The use of labile trivacant Keggin ions instead of relatively stable monovacant POMs enabled adaptation of the POM geometry and nuclearity to the Tc^{IV}_2 -units, which in turn enforced asymmetry of the POM within the crystalline lattice, enabling structure description without disorder. However, the labile POMs challenged obtaining pure phases, which we will address in the future by translating this and similar chemistry to organic media *via* alkylammonium counter cation use.

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Conflicts of interest

There are no conflicts to declare.

Data availability

The data supporting this article have been included as part of the supplementary information (SI). Supplementary information: 23 additional works are cited in the supporting information.²⁶ See DOI: 10.1039/d5cc06618b.

CCDC 2500437–2500439 contain the supplementary crystallographic data for this paper.^{27a–c}

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